

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 2nd Semester B.Sc. Examination, 2020
CSE – 1202, Digital Logic Design

Total Marks: 70

Time: 3 Hours

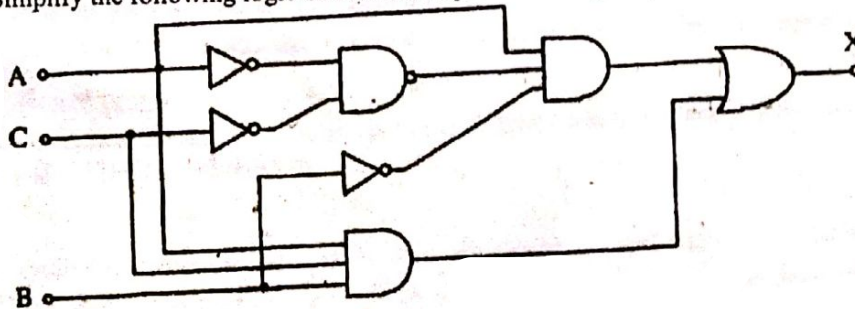
(Answer any 5 (Five) of the following Questions)

1. a) Define fan-out and propagation delay. 2
 b) Show how a two-input NAND gate can be constructed using two-input NOR gate? 4
 c) Simplify the Boolean function to a minimum number of literals and draw the diagram. 4
 I. $xyz + x'z + yz + x'y$
 II. $(x' + y)(x + z) + xz$
 d) Simplify the Boolean function with Karnaugh map $F(w,x,y,z) = \sum(1,3,7,11,15)$ and don't care conditions $d(w,x,y,z) = \sum(0,2,5)$. 4

2. a) What is the counter modulus? 1
 b) Describe with the figure a JK flip-flop. what is the benefit of using a JK flip-flop instead of an SR flip-flop? 5
 c) Implement a 4-bit synchronous counter and explain its functions. 4
 d) A 4-bit asynchronous ripple counter starts off in the 0000 state and then clock pulses are applied. Sometime later the clock pulses are removed and counter FFS read 0100. How many clock pulses have occurred? 4

3. a) What is fundamental mode operation? Explain asynchronous sequential circuit block diagram. 4
 b) Differentiate between Mealy and Moore's machines. 2
 c) Explain pulse mode logic with a block diagram. 3
 d) What is hazard? Design of (SIC) hazards free network with the following function $F(x,y,z) = \sum(1,3,5,7)$ 5

4. a) Find the 9's complement of $(25.639)_{10}$ 2
 b) Prove the universality of NAND and NOR gates. 4
 c) Develop a truth table for the following SOP expression: 3
 $A'B + ABC' + A'C' + AC$
 d) Simplify the following logic circuit and implement using only NAND gates. 5



5.
 - a) Draw the circuit diagram of a full adder using half adders and OR gate. 2
 - b) Write the output expressions and draw the circuit diagram of a 3-bit Comparator. 5
 - c) Write the output expressions of a Decimal to BCD priority encoder. 3
 - d) Draw the block diagram with the truth table of a 4x16 decoder using two 3x8 decoders. 4

6.
 - a) How does a priority encoder differ from an ordinary encoder? 2
 - b) Draw the block diagram and truth table of a BCD to 7 segment display system. 4
 - c) Implement a full adder circuit with a decoder and two OR gates. 3
 - d) Implement the function using an 8x1 Multiplexer. $F(A,B,C,D) = \sum m(0,2,4,5,6,8,10,13)$ 5

7.
 - a) Define fan-out, propagation delay and noise margin. 2
 - b) Simplify the following function using Quine-McCluskey method: 6

$$F(w,x,y,z) = \sum m(0,1,3,7,8,9,11,15)$$
 - c) Implement the following functions using PLA: 6

$$F_1(x, y, z) = \sum m(1, 2, 4, 6)$$

$$F_2(x, y, z) = \sum m(0, 1, 6, 7)$$

$$F_3(x, y, z) = \sum m(2, 6)$$

$$F_4(x, y, z) = \sum m(1, 2, 3, 5, 7)$$

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PHY – 1203, Physics

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(Answer any 5 (Five) of the following Questions)

1. a) What do you mean by entropy? Prove that the entropy increases at irreversible process. 3
 b) Mention some conditions of adiabatic change. Prove that $TV^{\gamma-1} = \text{Constant}$. 7
 c) At 0°C temperature, a fixed amount of gas is expanded double in volume. If $\gamma = 1.4$ then find out the last temperature. 4

2. a) Describe the first law of thermodynamics. Show that, $dw = PdV$. 3
 b) Describe each stage of Carnot's cycle and find out its efficiency. 7
 c) 40 gms of Hydrogen gas at 27°C are isothermally compressed to one fourth of the original volume. Find the value of work done during the process. Here $R = 8.4 \text{ J/gm-mole}$. 4

3. a) Define periodic motion and damped vibrations. 3
 b) Show that the total energy of the system is the same as the maximum value of any one of the two forms of energy. 8
 c) A spring hung vertically, is found to be stretched by 0.02m from its equilibrium position when a force of 4N acts on it. Then a 2 kg body is attached to the end of the spring and is pulled 0.04m from its equilibrium positions along the vertical line. What is the mechanical energy of the oscillating system? 3

4. a) What Simple Harmonic Oscillation? Write down its characteristics. 3
 b) Show that the average kinetic energy and average potential energy is same and constant and half of total energy. 7
 c) A body describing SHM has a maximum acceleration of $8\pi \text{ m/s}^2$ and a maximum speed of 1.6m/s. Find the time period T and the amplitude a. 4

5. a) Write short notes on (i) interferometer (ii) resolving power 3
 b) Show that interference obeys the conservation principle of light. 7
 c) Green light of wavelength $5100 \text{ \AA}$ from a narrow slit is incident on a double slit. If the overall separation of 10 fringes on a screen, 200cm, away is 2cm, find the slit separation. 4

6. a) What is packing fraction? Why the packing fraction value is less than 1? Mention which structure has large value of packing fraction. 3
 b) What is a Miller index? Show that for cubic crystal structure the interplanar distance $d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$. 7
 c) The lattice constant of a simple lattice is a. find the lattice spacing between (111), (112) and (113) lattice planes. 4

7. a) Define (i) Newton's ring (ii) Coherent source (iii) Retardation plates 3
 b) Describe the Brewster's Law. From Brewster's law show that the polarizing angle of incidence, the reflected and refracted rays are 90° apart. 7
 c) Light from a sodium vapour lamp ($\lambda = 0.58 \mu\text{m}$) forms an interference pattern on a screen 1.00m from a pair of slits. The bright fringes in pattern are 30mm apart. What is the slit separation? 4

MATH – 1204, Method of integration, differential equations & Series

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) Define Order and Degree of a differential equation.
b) Find the differential equation of all straight lines at a fixed distance P from the origin.
c) Solve (any 03) i. $\left(\frac{dy}{dx}\right) = x + y$ ii. $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$
iii. $y(1+xy)dx - xdy = 0$ iv. $\log\left(\frac{dy}{dx}\right) = ax + by$
2. a) Describe linearity with an example.
b) Solve the IVP $\frac{dy}{dx} + 2xy = f(x), y(0) = 2$ where $f(x) = \begin{cases} x, & 0 \leq x < 1 \\ 0, & x \geq 1 \end{cases}$
c) State Euler's method. Using Euler's method, obtain an approximation of $y(2.5)$ using first $h=0.1$ and then $h=0.05$ for IVP $y' = 0.1\sqrt{y} + 0.4x^2, y(2) = 4$
3. a) What do you mean by the surface of revolution? Explain the surface area of a solid revolution.
b) Find the length of the Arc of the curve $p^2 = ar$; from $r = a$ to $r = 2a$
c) Find the surface area of the solid generated by revolving the cycloid $x = a(\theta + \sin \theta), y = a(1 - \cos \theta)$ about its base.
4. a) Describe Simpson's rule with a graph.
b) Show that the series $\sum_{n=1}^{\infty} \frac{4.7.10 \dots (3n+1)}{1.2.3 \dots n} x^n$ converges if $x < -\frac{1}{3}$ and diverges if $x \geq \frac{1}{3}$.
c) Evaluate $\int_a^\beta \sqrt{(x-\alpha)(\beta-x)} dx$.
5. Evaluate
(i) $\int e^{2x} \left(\frac{1 + \sin 2x}{1 + \cos 2x} \right) dx$ (ii) $\int \frac{2x^2 - 1}{(x+1)^2(x-2)} dx$
(iii) $\int x \sqrt{\frac{1-x^2}{1+x^2}} dx$ (iv) $\int (\sin^{-1} x)^2 dx$

$$3.5x^4 = 14$$

6. a) State and prove Taylor's theorem with the remainder. 2+5=7
- b) Find nth Taylor's polynomial for $\ln x$ in powers of $(x - 2)$. 5
- c) Write down the failures of Taylor's series. 2
7. a) The population of a community is known to increase at a rate proportional to the number of people present at time t . If the population has doubled in 50 years, how long will it take to triple? 4
- b) Expand $e^{\sin x}$ in Maclaurin's series. 5
- c) Show that the arc length of the parabola $r(1 + \cos\theta) = 2$ from $\theta = 0$ to $\theta = \frac{\pi}{2}$ 5
 is $\sqrt{2} + \ln(1 + \sqrt{2})$

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ENG – 1205, Developing English Language Skills

Total Marks: 70

Time: 3 Hours

Part A (Comprehension: 20 marks)

Read the following passage and answer the questions given below:

There was a time when the study of literature and philosophy formed the basis of education. Knowledge of science gradually became indispensable in man's life. The question then arose, should literature continue to occupy its privileged position in our scheme of education in college and universities? There is no doubt that in the modern world we cannot think of doing without science. No one, therefore, would object to giving to science a place of prime importance in our curricula of studies. But that should in no way affect the position of literature. It is something as important, and there is no inherent enmity between the two.

Literature deals imaginatively with the stuff of which life is made. The study of what the men of letters have thought and felt has permanent interest for men. Literature gives us the unique experience of being in contact with a distinct personality which expresses itself through some "familiar matter" of real life or some fantasy. Science, on the other hand, deals with impersonal generalisation of universal principles. Literature studies from a personal point of view. It humanises every fact of life.

1. Answer the following questions: 1×5=5
 - (a) What does literature deal with?
 - (b) What, according to the passage, is the basis of education?
 - (c) What is the function of literature?
 - (d) What is the theme of the passage?
 - (e) What type of relation does the passage instigate in the passage?
2. Write the meaning of the following words in English. 1×5=5
 - (a) indispensable; (b) privileged; (c) distinct; (d) enmity; (e) fantasy.
3. Write the précis of the given passage. 5

Part B (Grammar: 25 marks)

4. Correct *any five* of the following sentences. 1×5=5
 - (a) I'm not in the habit to take heavy luncheon.
 - (b) It was the first they had have.
 - (c) I look forward to receive your reply.
 - (d) I can't but to laugh at this.
 - (e) The cost daily necessities are very high.
 - (f) I feel like to cry.
 - (g) A man is known by the company he is kept.
5. Fill in the gaps with the correct form of verbs: 0.5×10
= 5

Parents (a) ____ (be) the blessings of God to us. When a child (b) ____ (bear), their joys (c) ____ (know) no bounds. They (d) ____ (start) (e) ____ (think) of (f) ____ (nurse) and (g) ____ (bring) up their children. They (h) ____ (remain) safe in the custody of their parents since birth. Parents never mind (i) ____ (take) pains for the upbringing of their off-spring. We never (j) ____ (offend) our parents.

6. Fill in the gaps with appropriate articles:

0.5×10
= 5

There is a saying that sincerity is (a) ___ key to success. (b) ___ person can prosper in life by doing hard work. The man who does not follow (c) ___ rules of sincerity can never go (d) ___ long way in (e) ___ world. Many (f) ___ man is not conscious of (g) ___ importance of (h) ___ sincerity for which they don't have (i) ___ benefit of (j) ___ success. However, we should be sincere to our work if we want to achieve anything notable.

7. Fill in the gaps with appropriate prepositions:

0.5×10
= 5

We want happiness (a) ___ life. No one wants to live an unhappy life. It depends greatly (b) ___ our activities and attitude (c) ___ life. If we are dishonest and do evil activities, we will never be able to lead a happy life. Again, if we always hanker (d) ___ money or wealth, happiness will never be our company. (e) ___ the other hand, those who are honest, pious and satisfied (f) ___ what they have can enjoy happiness. Honest people never suffer from anxiety or fear while dishonest people remain (g) ___ anxiety or fear while dishonest people remain (h) ___ great fear and anxiety. Pious people always live (i) ___ incessant peace. They are not afraid (j) ___ anything. And those who are not covetous for money or wealth can enjoy pure happiness.

8. Transform *any five* of the following sentences as directed.

1×5=5

- (a) I called him but he didn't respond. (Make it Complex)
- (b) It's a very nice painting. (Make it Exclamatory)
- (c) Though he tried hard, he could not reach his goal. (Make it Compound)
- (d) He speaks more correctly than his brother. (Make it Positive Degree)
- (e) He made me deliver a speech in the meeting. (Make it Passive)
- (f) He is trying to survive well. (Make it Complex)
- (g) Sacrificing the dearest possession, Della proved her unfeigned love. (Make it Compound)

Part C: Writing (30 marks)

- 9. Write a paragraph on "The safety measures of minimizing the infections of the Corona Virus" 8
- 10. Write an application to the Principal/Head of your college/Institute for the arrangement of a debate competition. 7
- 11. Amplify the idea "Look before you leap". 8
- 12. Compose an Email to your bosom friend to celebrate his/her birthday. 7

- c) What will be the output of the following program when run? 3
- ```

int main(){
 int a=20, b=5, *ptr;
 ptr=&a;
 b=*ptr;
 printf(" %d= %d= %d", ++a, b++, *ptr);
 return 0;
}

```
- d) Show the memory representation of the following declaration 4
- ```

int m=0, n=20;
int *ptr=&m;
n=*ptr;

```
4. a) What is recursion? 1+4=5
- A function named fact(int) is given below. Trace the call fact(5). Show how you reached the return value of this call by drawing a function call tree.
- ```

int fact(int){
 if (n==0 || n==1)
 return 1;
 else
 return (n*fact(n-1))
}

```
- b) What is dynamic memory allocation? Distinguish between malloc() and calloc(). 2+2=4
- c) Describe the limitations of using getchar and scanf functions for reading string. 2
- d) Difference between else\_if ladder and switch statement. 3
5. a) Explain creating, opening and closing a data file in c with example. 4
- b) What is the difference between file opening mode r+ and w+ in c? 2
- c) What is the importance of file closing? 2
- d) A file named sorting.text contains a series of integer numbers. Write a program that read an array and after sorting these number in ascending order save a result in an existing file. 6
6. a) Draw the flowcharts with appropriate syntax for the following statements: 6
- Nested if...else statement
  - switch statement
  - while statement
- b) Given a mark ranging from 0 to 100 of a student named Karim, write a program to compute and print the Grade for the student. 3
- | Range     | Grade |
|-----------|-------|
| 80 to 100 | A+    |
| 71 to 79  | A     |
| 61 to 70  | B     |
| 51 to 60  | C     |
| 40 to 50  | D     |
| < 40      | F     |
- c) Write a for statement to print/compute the following sequence of integers: 2x2.5=5
- $5 + 10 + 15 + \dots + 500$
  - 1, 3, 9, 27, 81, 243
7. a) Define Array. Explain the declaration of various kinds of Arrays. 2+3=5
- b) What is a data structure? Why is an array called a data structure? 2+2=4
- c) Write a c program to print the elements of an array in reverse order. 5